



UNIVERSITY OF THE PUNJAB

Seventh Semester – 2019

Examination: B.S. 4 Years Program

Roll No.

PAPER: Basic Econometrics (Theory)
Course Code: STAT-403 Part – II

MAX. TIME: 2 Hrs. 45 Min.
MAX. MARKS: 50

ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED

Q.2. Answer to the following short questions.

(4x5=20)

- (i) Methodology of Econometric research.
- (ii) Assumptions of M.L. estimation for GL regression
- (iii) Ridge Regression
- (iv) Test-statistic to test some regression co-efficients in GL regression
- (v) Sampling distribution of regression sum of squares..

Long questions

Q.3. For S.L. Regression $Y = \alpha + \beta X + \epsilon$, show that ML estimate of α is BLUE. (07)

Q.4. Consider the partitioned G.L. regression $\underline{Y} = X_1 \underline{\beta}_1 + X_2 \underline{\beta}_2 + \underline{\epsilon}$. Show that lower right-hand block in $(X' X)^{-1}$ may be expressed as $(X_2' M_1 X_2)^{-1}$, where $M_1 = I - X_1 (X_1' X_1)^{-1} X_1'$. Give a least squares interpretation of $M_1 X_2$ and $X_2' M_1 X_2$. (07)

Q.5. A production function model is specified as $Y = \beta_1 + \beta_2 X_2 + \beta_3 X_3 + u$ (09)

The data refer to a sample of 23 firms and following calculations are made as deviations from the sample means.

$$\sum x_2^2 = 12, \quad \sum x_3^2 = 12, \quad \sum y^2 = 10, \quad \sum x_2 x_3 = 8, \quad \sum x_2 y = 10, \quad \sum x_3 y = 8$$

What is the L.S. estimate of β_3 and its standard error, when $\beta_2 + \beta_3 = 1$

Q.6. Consider the partitioned G.L. regression $\underline{Y} = X_1 \underline{\beta}_1 + X_2 \underline{\beta}_2 + \underline{\epsilon}$. Obtain stepwise regression estimates of $\underline{\beta}_1$ and $\underline{\beta}_2$. Compare these estimates with full regression estimates. (07)



UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program : Seventh Semester – Fall 2021

Paper: Basic Econometrics (Theory)

Course Code: STAT-403

Roll No.

Time: 3 Hrs. Marks: 60

Q.1. Answer the following short questions: (6x5=30)

- Explain the use of extraneous information in linear regression.
- Discuss the methodology of Econometric research.
- Under what reasons an error term is introduced in Econometric model?
- Discuss Restricted Least Squares method.
- Write short note on Stepwise regression.
- Discuss the assumptions of MLE method for GLR model.

Answer the following questions. (3x10=30)

Q.2. Show that least squares unbiased estimate of variance of disturbance term is given by

$$\hat{\sigma}_u^2 = \frac{\sum e_i^2}{n - k - 1}$$

Where e_i the i th residual, k is the number of explanatory variables of the GLR model

Q.3. Consider the General linear regression model $\underline{Y} = \underline{X}\underline{\beta} + \underline{\varepsilon}$ which fulfills the OLS assumptions. Develop the testing procedure for $H_0: \underline{\beta} = \underline{\beta}_0$.

Q.4. For the Simple linear regression model, show that the maximum likelihood and OLS estimators of $\underline{\beta}$ are equal.

PUACP



UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Seventh Semester – Fall 2022

Paper: Basic Econometrics (Theory)

Course Code: STAT-403

Roll No.

Time: 3 Hrs. Marks: 60

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Answer the following short questions:

(5x6=30)

- Assumptions of simple linear regression model.
- Methodology of Econometric research.
- Importance of Gauss Markov theorem.
- Stepwise regression.
- Need of econometrics as a separate discipline.

Answer the following questions.

(3x10=30)

- Q.2. Discuss the sampling distributions of error and total sum of squares used in regression analysis when there is only one regressor and model obeys the assumptions of OLS.
- Q.3. Let the GLR model be partitioned as:

$$\underline{Y} = X_1\underline{\beta}_1 + X_2\underline{\beta}_2 + \underline{\epsilon}$$

Obtain the LS estimates of $\underline{\beta}_1$ and $\underline{\beta}_2$.

- Q.4. A production function model is specified as:

$$Y_i = \beta_1 + \beta_2 X_{2i} + \beta_3 X_{3i} + U_i$$

Where Y_i = log output, X_{2i} = log labor input, X_{3i} = log capital input. The data refer to a sample of 23 firms and observations are measured as deviations from the sample means.

$$\sum x_{2i}^2 = 12, \sum x_{2i}x_{3i} = 8, \sum x_{3i}^2 = 12, \sum x_{2i}y_i = 10, \sum y_i^2 = 10, \sum x_{3i}y_i = 8$$

Test the hypothesis that $H_0: \beta_2 + \beta_3 = 1$.



UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Seventh Semester – Fall 2024

Paper: Basic Econometrics (Theory)

Course Code: STAT-403

Roll No.

Marks: 60

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Answer the following short questions: (6x5=30)

- Major data type used in econometric.
- Find the covariance between the $\hat{\beta}_0$ and $\hat{\beta}_1$ in simple linear regression model.
- What are the economic meanings of "zero restrictions" on the parameter of model?
- Justify the inclusion of error term in econometric model.
- Find mean and variance of estimated variance of the disturbance term for SLR Model.
- Methodology of econometric research.

Answer the following questions. (3x10=30)

- Q.2.** Find means and variances of stepwise regression estimates. Also find covariance between these estimates.
- Q.3.** Prove that in multiple linear regression, the linear combination of least square estimators is BLUE of its corresponding linear function of parameters.

Q.4. A product function model is specified as;

$$Y_i = \beta_1 + \beta_2 X_{2i} + \beta_3 X_{3i} + \varepsilon_i$$

Where Y_i = log output, X_{2i} = log labor input, X_{3i} = log capital input.

The data refer to a sample of 23 firms and observations are measured as deviation from the sample means.

$$\sum x_{2i}^2 = 12, \sum x_{2i} x_{3i} = 8, \sum x_{3i}^2 = 12, \sum x_{2i} y_i = 10, \sum x_{3i} y_i = 8, \sum y_i^2 = 10$$

- Estimate β_2 and β_3 , their standard errors and R^2 .
- Test the hypothesis that $\beta_2 + \beta_3 = 1$.
- Suppose that you wish to impose a priori restriction that $\beta_2 + \beta_3 = 1$. What is the least squares estimate of β_2 and its S.E? what is the value of R^2 in this case. Compare the results obtained in (a) and comment.



UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program : Seventh Semester – 2021

Paper: Basic Econometrics (Theory)

Course Code: STAT-403

Part – II

Roll No.

021621

Time: 2 Hrs. 45 Min. Marks: 50

ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED

Q.2. Give short answers of the following:

(4x5=20)

- What is econometrics? Discuss its different stages of econometric research.
- What do you know about stepwise regression?
- Briefly discuss the underlying assumption of classical linear regression model.
- Explain the use of extraneous information in linear regression.

Answers the following questions.

(3x10=30)

Q.3. Show that the regression, correlation and ANOVA methods are equivalent to test the basic hypothesis that there is no linear relationship between the variables.

(10)

Q.4. Consider the General linear regression model $Y = X\beta + \varepsilon$ with no constant term and all the classical assumptions hold. Develop the test statistics to test the hypothesis $H_0: \beta = \beta_0$

(10)

Q.5. For the Simple linear regression model, obtain the maximum likelihood estimators of regression coefficient α and β .

(10)

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UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Seventh Semester – Fall 2023

Paper: Basic Econometrics (Theory)

Course Code: STAT-403

(Common with BS-AD 7th)

Roll No. 049133....

Time: 3 Hrs. Marks: 80

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Answer the following short questions:

(5x6=30)

- Assumptions of MLE method.
- Why stochastic models are used instead of mathematical model?
- Find the OLS estimator of α .
- Procedure of CHOW Test for K variable regression model.
- Restricted Least Square estimators.

Answer the following questions.

(3x10=30)

Q.2. Let the General linear regression model be partitioned as:

$$\underline{Y} = X_1\underline{\beta}_1 + X_2\underline{\beta}_2 + \underline{\epsilon}$$

Develop the test statistics to test the hypothesis $H_0: \underline{\beta}_2 = 0$

Q.3. Discuss the sampling distribution of total sum of squares and Regression sum of squares when there is only one regressor and model obeys the assumptions of OLS.

Q.4. The following estimated equation was made by OLS, using quarterly data for 1968 to 1986 inclusive.

$$\hat{Y}_t = 2.20 + 0.104X_{t1} - 3.48X_{t2} + 0.34X_{t3}$$

(3.4) (0.005) (2.2) (0.15)

(Standard errors are given in parenthesis)

The explained SS and residual SS was 109.6 and 18.48 respectively. Test the significance of each of the slope coefficients.



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Seventh Semester 2017
Examination: B.S. 4 Years Programme

Roll No.

PAPER: Basic Econometrics (Theory)
Course Code: STAT-403

TIME ALLOWED: 2 hrs. & 30 mins.
MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.

Q.2. Discuss the following.

(20)

- (i) The properties of OLS estimators
- (ii) Stepwise regression
- (iii) CHOW test
- (iv) Functions of Econometrics
- (v) Extraneous information's and its use.

Q.3. For GLR model $\underline{Y} = X_1\underline{\beta}_1 + X_2\underline{\beta}_2 + \underline{u}$, obtain least squares estimates of $\underline{\beta}_1$ and $\underline{\beta}_2$. (06)

Q.4. For GLR model $\underline{Y} = X\underline{\beta} + \underline{\epsilon}$, show that ML estimator of $\underline{\beta}$ is BLUE. (06)

Q.5. The following calculations are made from 10 sets of observations on Y, X_1 and X_2 which follow (06)

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + u$$

$$\Sigma Y = 20, \quad \Sigma X_1 = 30, \quad \Sigma X_2 = 40, \quad \Sigma X^2 = 88.2, \quad \Sigma X_1^2 = 92, \quad \Sigma X_2^2 = 163,$$

$$\Sigma X_1 Y = 59, \quad \Sigma X_2 Y = 88, \quad \Sigma X_1 X_2 = 119$$

Test the significance of each regression Co-efficient.

Q.6. Consider the partitioned model $\underline{Y} = X_1\underline{\beta}_1 + X_2\underline{\beta}_2 + \underline{u}$ let b_1 and b_2 be the OLS estimators of β_1 and β_2 respectively. Suppose that $E\underline{u} = X_2\lambda$. Check the unbiasedness of b_1 and b_2 . (06)

Q.7. Consider SLR $Y = \alpha + \beta X + \epsilon$, Show that $\bar{Y}$ and LS estimator of β are independent. (06)



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Attempt this Paper on Separate Answer Sheet provided.

Q.2. Discuss the following.

(20)

- (i) Properties of MLE of error variance in G.L. Regression.
- (ii) Stepwise regression
- (iii) Gauss Markov's theorem
- (iv) Functions of Econometrics
- (v) Restricted L.S. estimation

Q.3. For GL Regression $\underline{Y} = X\underline{\beta} + \underline{\epsilon}$, develop the testing procedure to test some regression co-efficient (not all). (08)

Q.4. Parameters of the model $\underline{Y} = X\underline{\beta} + \underline{\epsilon}$ follow the linear relations $R\underline{\beta} = \underline{\gamma}$. Obtain least square estimate of $\underline{\beta}$ under the given relation. Also find variance covariance matrix of the derived estimators. (08)

Q.5. We are given the following calculations such that variables follow the relation (08)

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \epsilon$$

$$n = 5, \quad \Sigma X_1 = 15, \quad \Sigma X_2 = 25, \quad \Sigma Y = 20, \quad \Sigma X_1^2 = 55, \quad \Sigma X_2^2 = 129, \\ \Sigma Y^2 = 108, \quad \Sigma X_1 Y = 76, \quad \Sigma X_2 Y = 109, \quad \Sigma X_1 X_2 = 81$$

Test the hypothesis that Co-efficient of X_1 and X_2 are equal in magnitude but opposite in sign.

Q.6. If the sample values of X in the linear model $Y = \alpha + \beta X + u$, have zero mean, show that the covariance of the least squares estimates of α and β is zero. (06)